Homework 10
Elementary Linear Algebra
(MATH 2250-03), Spring, 2019

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Section 6.1 - Inner Product, Length, and Orthogonality

1. Let $u = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ and $v = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$. Compute the following quantities:

(a) $\|u\| - \|v\|$

(b) $\|u - v\|$

(c) $u \cdot u$

(d) $v \cdot u$

(e) $\left(\frac{u \cdot v}{v \cdot v}\right) v$

(f) $\left(\frac{1}{u \cdot u}\right) u$

Using the definition of the dot product and the identity $\|x\| = \sqrt{x \cdot x}$, we may compute our desired quantities as follows:

(a)

$$\begin{aligned} \|u\| - \|v\| &= \sqrt{u \cdot u} - \sqrt{v \cdot v} \\ &= \sqrt{(-1)^2 + (2)^2} - \sqrt{(4)^2 + (6)^2} \\ &= \sqrt{5} - \sqrt{52} \\ &= \sqrt{5} - 2\sqrt{13} \end{aligned}$$

(b)

$$\begin{aligned} \|u - v\| &= \sqrt{(u - v) \cdot (u - v)} \\ &= \sqrt{\big((-1) - (4)\big)^2 + \big((2) - (6)\big)^2} \\ &= \sqrt{25 + 16} \\ &= \sqrt{41} \end{aligned}$$

(c)

$$\begin{aligned} u \cdot u &= (-1)^2 + (2)^2 \\ &= 1 + 4 \\ &= 5 \end{aligned}$$

(c)

$$\begin{aligned} v \cdot u &= (4)(-1) + (6)(2) \\ &= -4 + 12 \\ &= 8 \end{aligned}$$

(d)

$$\begin{aligned} \left(\frac{u \cdot v}{v \cdot v}\right) v &= \frac{(-1)(4) + (2)(6)}{(4)^2 + (6)^2} \begin{bmatrix} 4 \\ 6 \end{bmatrix} \\ &= \frac{8}{52} \begin{bmatrix} 4 \\ 6 \end{bmatrix} \\ &= \begin{bmatrix} 8/13 \\ 12/13 \end{bmatrix} \end{aligned}$$

(e)

$$\begin{aligned} \left(\frac{1}{u \cdot u}\right) u &= \frac{1}{(-1)^2 + (2)^2} \begin{bmatrix} -1 \\ 2 \end{bmatrix} \\ &= \frac{1}{5} \begin{bmatrix} -1 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} -1/5 \\ 2/5 \end{bmatrix} \end{aligned}$$

2. Find the distance between $x = \begin{bmatrix} 10 \\ -3 \end{bmatrix}$ and $y = \begin{bmatrix} -1 \\ -5 \end{bmatrix}$.

By definition, the distance between x and y, denoted $d(x,y)$, is given by $d(x,y) = \|x-y\|$. Using this, we compute the distance between x and y with the following calculation:

$$\begin{aligned} d(x,y) &= \|x-y\| \\ &= \sqrt{(x-y)\cdot(x-y)} \\ &= \sqrt{\big((10)-(-1)\big)^2 + \big((-3)-(-5)\big)^2} \\ &= \sqrt{(11)^2 + (2)^2} \\ &= 5\sqrt{5}. \end{aligned}$$

3. Let W be a subspace of $\mathbb{R}^n$. Show that if $x \in \mathbb{R}^n$ is a vector which satisfies both $x \in W$ and $x \in W^\perp$, then $x = 0$.

Suppose W is a subspace of $\mathbb{R}^n$ and $x \in \mathbb{R}^n$ is a vector which satisfies both $x \in W$ and $x \in W^\perp$. Recall that, by definition, $W^\perp$ is the subspace of $\mathbb{R}^n$ defined by

$$W^\perp = \left\{ v \in \mathbb{R}^n : v \cdot w = 0 \ \text{ for all } \ w \in W \right\}.$$

So, if $x \in W^\perp$, then $x \cdot w = 0$ for all $w \in W$. But, since we have also supposed that $x \in W$ as well, this yields that x must satisfy

$$x \cdot x = 0.$$

By Theorem 1 of Section 6.1 in your textbook, a vector $u \in \mathbb{R}^n$ satisfies $u \cdot u = 0$ if and only if $u = 0$. Therefore, we may conclude that if W is a subspace of $\mathbb{R}^n$ and $x \in \mathbb{R}^n$ satisfies both $x \in W$ and $x \in W^\perp$, then $x = 0$.

4. Determine whether

$$\left\{ \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix}, \begin{bmatrix} -1/2 \\ 0 \\ 1/2 \end{bmatrix} \right\}$$

is an orthogonal set. If it is, normalize the vectors in this set to produce an orthonormal set.

Recall that, by definition, a set of vectors $\{u_1, \ldots, u_p\}$ in $\mathbb{R}^n$ is an orthogonal set if $u_i \cdot u_j = 0$ whenever $i \neq j$ for all $i, j \in \{1, \ldots, p\}$.

So, to determine whether the set of two vectors in $\mathbb{R}^3$ given in the statement of the problem is an orthogonal set, we need only compute their dot product:

$$\begin{aligned} \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} \cdot \begin{bmatrix} -1/2 \\ 0 \\ 1/2 \end{bmatrix} &= (1/3)(-1/2) + (1/3)(0) + (1/3)(1/2) \\ &= -1/6 + 1/6 \\ &= 0. \end{aligned}$$

As the dot product computed above is indeed zero, we may conclude that $\left\{ \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix}, \begin{bmatrix} -1/2 \\ 0 \\ 1/2 \end{bmatrix} \right\}$ is an orthogonal set.

To normalize this set of vectors, we need only multiply each vector by the reciprocal of its length. That is, to normalize a vector $u \in \mathbb{R}^n$, we simply multiply u by the scalar $\dfrac{1}{\|u\|}$. As such, we proceed with the following calculations:

$$\left(\frac{1}{\left\| \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} \right\|} \right) \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} = \left(\frac{1}{\sqrt{3(1/3)^2}} \right) \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} = \sqrt{3} \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} = \begin{bmatrix} \sqrt{3}/3 \\ \sqrt{3}/3 \\ \sqrt{3}/3 \end{bmatrix}.$$

$$\left(\frac{1}{\left\| \begin{bmatrix} -1/2 \\ 0 \\ 1/2 \end{bmatrix} \right\|} \right) \begin{bmatrix} -1/2 \\ 0 \\ 1/2 \end{bmatrix} = \left(\frac{1}{\sqrt{(-1/2)^2 + (1/2)^2}} \right) \begin{bmatrix} -1/2 \\ 0 \\ 1/2 \end{bmatrix} = \sqrt{2} \begin{bmatrix} -1/2 \\ 0 \\ 1/2 \end{bmatrix} = \begin{bmatrix} -\sqrt{2}/2 \\ 0 \\ \sqrt{2}/2 \end{bmatrix}.$$

As the above calculations have normalized the vectors $\begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix}$ and $\begin{bmatrix} -1/2 \\ 0 \\ 1/2 \end{bmatrix}$, we may conclude that $\left\{ \begin{bmatrix} \sqrt{3}/3 \\ \sqrt{3}/3 \\ \sqrt{3}/3 \end{bmatrix}, \begin{bmatrix} -\sqrt{2}/2 \\ 0 \\ \sqrt{2}/2 \end{bmatrix} \right\}$ is an orthonormal set.

5. Compute the orthogonal projection of $\begin{bmatrix} 1 \\ 7 \end{bmatrix}$ onto the line through $\begin{bmatrix} -4 \\ 2 \end{bmatrix}$ and the origin.

Recall that the orthogonal projection of a vector $x \in \mathbb{R}^n$ onto the line $L = \text{span}\{u\}$ through a nonzero vector $u \in \mathbb{R}^n$ and the origin, denoted $\text{proj}_L(x)$ (or, if you prefer, denoted by $\text{proj}_u(x)$ instead) is given by

$$\text{proj}_L(x) = \frac{x \cdot u}{u \cdot u} u.$$

Hence, we may compute the orthogonal projection of $x = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$ onto the line $L = \text{span}\{u\}$ through $u = \begin{bmatrix} -4 \\ 2 \end{bmatrix}$ and the origin as follows:

$$\begin{aligned} \text{proj}_L(x) = \frac{x \cdot u}{u \cdot u} u &= \frac{(1)(-4) + (7)(2)}{(-4)^2 + (2)^2} \begin{bmatrix} -4 \\ 2 \end{bmatrix} \\ &= \frac{-4 + 14}{16 + 4} \begin{bmatrix} -4 \\ 2 \end{bmatrix} \\ &= \frac{10}{20} \begin{bmatrix} -4 \\ 2 \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} -4 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} -2 \\ 1 \end{bmatrix}. \end{aligned}$$

6. Given a vector $u \neq 0$ in $\mathbb{R}^n$, let $\mathrm{L} = \operatorname{span}\{u\}$. Show that the mapping $x \mapsto \operatorname{proj}_{\mathrm{L}}(x)$ is a linear transformation.

As stated in Question 5, recall that the orthogonal projection of a vector $x \in \mathbb{R}^n$ onto the line $\mathrm{L} = \operatorname{span}\{u\}$ through a vector $u \in \mathbb{R}^n$ and the origin, denoted $\operatorname{proj}_{\mathrm{L}}(x)$ is given by

$$\operatorname{proj}_{\mathrm{L}}(x) = \frac{x \cdot u}{u \cdot u} u.$$

So, to show the mapping which sends $x \in \mathbb{R}^n$ to $\operatorname{proj}_{\mathrm{L}}(x)$ is a linear transformation, we need only show that $\operatorname{proj}_{\mathrm{L}}(cx) = c \ \operatorname{proj}_{\mathrm{L}}(x)$ and $\operatorname{proj}_{\mathrm{L}}(x+y) = \operatorname{proj}_{\mathrm{L}}(x) + \operatorname{proj}_{\mathrm{L}}(y)$ for any scalar $c \in \mathbb{R}$ and any vectors $x, y \in \mathbb{R}^n$. As such, note that if $c \in \mathbb{R}$ is any scalar and $x, y \in \mathbb{R}^n$ are any vectors, the properties of the dot product described in Theorem 1 of Section 6.1 of your textbook yield that:

$$\begin{aligned}
\operatorname{proj}_{\mathrm{L}}(cx) &= \frac{(cx) \cdot u}{u \cdot u} u \\
&= \frac{c(x \cdot u)}{u \cdot u} u \\
&= c\left(\frac{x \cdot u}{u \cdot u} u\right) \\
&= c \ \operatorname{proj}_{\mathrm{L}}(x).
\end{aligned}$$

$$\begin{aligned}
\operatorname{proj}_{\mathrm{L}}(x+y) &= \frac{(x+y) \cdot u}{u \cdot u} u \\
&= \frac{(x \cdot u) + (y \cdot u)}{u \cdot u} u \\
&= \left(\frac{x \cdot u}{u \cdot u} + \frac{y \cdot u}{u \cdot u}\right) u \\
&= \left(\frac{x \cdot u}{u \cdot u} u\right) + \left(\frac{y \cdot u}{u \cdot u} u\right) \\
&= \operatorname{proj}_{\mathrm{L}}(x) + \operatorname{proj}_{\mathrm{L}}(y).
\end{aligned}$$

As we have shown that $\operatorname{proj}_{\mathrm{L}}(cx) = c \ \operatorname{proj}_{\mathrm{L}}(x)$ and $\operatorname{proj}_{\mathrm{L}}(x+y) = \operatorname{proj}_{\mathrm{L}}(x) + \operatorname{proj}_{\mathrm{L}}(y)$ for any scalar $c \in \mathbb{R}$ and any vectors $x, y \in \mathbb{R}^n$, we may conclude that the mapping $x \mapsto \operatorname{proj}_{\mathrm{L}}(x)$ is indeed a linear transformation.

7. Let $u_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $u_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ and $y = \begin{bmatrix} -1 \\ 4 \\ 3 \end{bmatrix}$. Verify that $\{u_1, u_2\}$ is an orthogonal set. Then, find the orthogonal projection of y onto $\text{span}\{u_1, u_2\}$.

To verify that $\{u_1, u_2\}$ is an orthogonal set, we proceed as in Question 4 and compute $u_1 \cdot u_2$ as follows:

$$u_1 \cdot u_2 = (1)(-1) + (1)(1) + (0)(0) = -1 + 1 = 0.$$

So, it is indeed the case that $\{u_1, u_2\}$ is an orthogonal set.

Following this, recall that the orthogonal projection of a vector $x \in \mathbb{R}^n$ onto a subspace W of $\mathbb{R}^n$ spanned by the orthogonal basis $\{u_1, \ldots, u_p\}$ is given by

$$\text{proj}_W(x) = \frac{x \cdot u_1}{u_1 \cdot u_1} u_1 + \ldots + \frac{x \cdot u_p}{u_p \cdot u_p} u_p.$$

So, to find the orthogonal projection of y onto $W = \text{span}\{u_1, u_2\}$, we proceed as follows:

$$\begin{aligned}
\text{proj}_W(y) &= \frac{y \cdot u_1}{u_1 \cdot u_1} u_1 + \frac{y \cdot u_2}{u_2 \cdot u_2} u_2 \\
&= \frac{(-1)(1) + (4)(1) + (3)(0)}{(1)^2 + (1)^2 + (0)^2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \frac{(-1)(-1) + (4)(1) + (3)(0)}{(-1)^2 + (1)^2 + (0)^2} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \\
&= \frac{3}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \frac{5}{2} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \\
&= \begin{bmatrix} 3/2 - 5/2 \\ 3/2 + 5/2 \\ 0 \end{bmatrix} \\
&= \begin{bmatrix} -1 \\ 4 \\ 0 \end{bmatrix}.
\end{aligned}$$

8. Let $v_1 = \begin{bmatrix} 2 \\ -1 \\ -3 \\ 1 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix}$, and $z = \begin{bmatrix} 3 \\ -7 \\ 2 \\ 3 \end{bmatrix}$. Let $W = \operatorname{span}\{v_1, v_2\}$ and find the best approximation to z by elements of W.

Recall that the Best Approximation Theorem (Theorem 9 of Section 6.3 in your textbook) states that for any subspace W of $\mathbb{R}^n$ and any vector $y \in \mathbb{R}^n$, the vector $\hat{y} = \operatorname{proj}_W(y)$ is the closest point in W to y in the sense that $\|y - \hat{y}\| < \|y - v\|$ for all vectors $v \in W$ satisfying $v \neq \hat{y}$.

As such, we know that the best approximation to z by elements of $W = \operatorname{span}\{v_1, v_2\}$ is given by $\operatorname{proj}_W(z)$. To compute $\operatorname{proj}_W(z)$, we must first confirm that $\{v_1, v_2\}$ is an orthogonal set. To do so, we simply compute

$$v_1 \cdot v_2 = (2)(1) + (-1)(1) + (-3)(0) + (1)(-1) = 2 - 1 + 0 - 1 = 0.$$

So, $\{v_1, v_2\}$ is indeed an orthogonal set. Moreover, as $W = \operatorname{span}\{v_1, v_2\}$, the set $\{v_1, v_2\}$ is also an orthogonal basis for W. Hence, we have $\operatorname{proj}_W(z) = \frac{z \cdot v_1}{v_1 \cdot v_1} v_1 + \frac{z \cdot v_2}{v_2 \cdot v_2} v_2$, and the best approximation to z by elements of W is given by

$$\begin{aligned}
\operatorname{proj}_W(z) &= \frac{z \cdot v_1}{v_1 \cdot v_1} v_1 + \frac{z \cdot v_2}{v_2 \cdot v_2} v_2 \\
&= \frac{(3)(2) + (-7)(-1) + (2)(-3) + (3)(1)}{(2)^2 + (-1)^2 + (-3)^2 + (1)^2} v_1 + \frac{(3)(1) + (-7)(1) + (2)(0) + (3)(-1)}{(1)^2 + (1)^2 + (0)^2 + (-1)^2} v_2 \\
&= \frac{2}{3} \begin{bmatrix} 2 \\ -1 \\ -3 \\ 1 \end{bmatrix} + \frac{-7}{3} \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix} \\
&= \begin{bmatrix} 4/3 \\ -2/3 \\ -6/3 \\ 2/3 \end{bmatrix} + \begin{bmatrix} -7/3 \\ -7/3 \\ 0 \\ 7/3 \end{bmatrix} \\
&= \begin{bmatrix} -1 \\ -3 \\ -2 \\ 3 \end{bmatrix}.
\end{aligned}$$

9. Let W be a subspace of $\mathbb{R}^n$ with an orthogonal basis $\{w_1, \ldots, w_p\}$. Suppose $\{v_1, \ldots, v_q\}$ is an orthogonal basis for $W^\perp$. Show that $\{w_1, \ldots, w_p, v_1, \ldots, v_q\}$ is an orthogonal basis for $\mathbb{R}^n$.

Recall that the Orthogonal Decomposition Theorem (Theorem 8 of Section 6.3 in your textbook) states that for any subspace W of $\mathbb{R}^n$ and any vector $y \in \mathbb{R}^n$, y can be written uniquely as $y = \hat{y} + z$, where $\hat{y} \in W$ and $z \in W^\perp$.

So, if W is a subspace of $\mathbb{R}^n$ with an orthogonal basis $\{w_1, \ldots, w_p\}$ and $\{v_1, \ldots, v_q\}$ is an orthogonal basis for $W^\perp$, then by writing $\hat{y} = c_1 w_1 + hdots c_p w_p$ for some constants $c_1, \ldots, c_p \in \mathbb{R}$ and $z = d_1 v_1 + \ldots + d_q v_q$ for some constants $d_1, \ldots, d_q$, the orthogonal decomposition theorem yields that every vector $y \in \mathbb{R}^n$ can be written uniquely as

$$y = \hat{y} + z = c_1 w_1 + \ldots + c_q w_q + d_1 v_1 + \ldots + d_q v_q.$$

So, we may conclude that $\{w_1, \ldots, w_p, v_1, \ldots, v_q\}$ spans $\mathbb{R}^n$.

To show that $\{w_1, \ldots, w_p, v_1, \ldots, v_q\}$ is an orthogonal basis for $\mathbb{R}^n$, we need only show that $\{w_1, \ldots, w_p, v_1, \ldots, v_q\}$ is both a linearly independent set of vectors and an orthogonal set. However, this can be made easier by considering Theorem 4 in Section 6.2 of your textbook, which states that any orthogonal set of nonzero vectors is a linearly independent set. That is, if we can show that $\{w_1, \ldots, w_p, v_1, \ldots, v_q\}$ is an orthogonal set of nonzero vectors, then it is also a linearly independent set and therefore (as we have already shown it spans $\mathbb{R}^n$) an orthogonal basis for $\mathbb{R}^n$.

So, to show that $\{w_1, \ldots, w_p, v_1, \ldots, v_q\}$ is an orthogonal set of nonzero vectors, first note that, as $\{w_1, \ldots, w_p\}$ is a basis for W and $\{v_1, \ldots, v_q\}$ is a basis for $W^\perp$, both $\{w_1, \ldots, w_p\}$ and $\{v_1, \ldots, v_q\}$ are linearly independent sets and cannot contain the zero vector. Moreover, since $w_1, \ldots, w_p$ is an orthogonal basis for W and $v_1, \ldots, v_q$ is an orthogonal basis for $W^\perp$, we have $v_i \cdot v_j = 0$ and $w_i \cdot w_j = 0$ for $i \neq j$. So, we need only show that $w_i \cdot v_j = 0$ for all $i \in \{1, \ldots, p\}$ and all $j \in \{1, \ldots, q\}$. However, this is easy to do, since every vector $v \in W^\perp$ satisfies $v \cdot w = 0$ for every vector $w \in W$ by definition. So, as $w_1, \ldots, w_p \in W$ and $v_1, \ldots, v_q \in W^\perp$, it is certainly the case that $\{w_1, \ldots, w_p, v_1, \ldots, v_q\}$ is an orthogonal set.

Therefore, as we have shown that $\{w_1, \ldots, w_p, v_1, \ldots, v_q\}$ is an orthogonal set of linearly independent vectors which spans $\mathbb{R}^n$, we may indeed conclude that $\{w_1, \ldots, w_p, v_1, \ldots, v_q\}$ is an orthogonal basis for $\mathbb{R}^n$.